UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #1

Course: **CHE 433. Process Simulation With Aspen Plus.**
Term: **Spring 2020**
Professor: **Dr. Michael Caracotsios**
Office: **EIB 244**
E-mail: **mcaracot@uic.edu**

Your name: Huda Usman
Number of points..............................100
Your score.................................

Note: **All problems are worth 100 points. Your final score will be the average value**

Problem 1: Review of Macroscopic Balances and Numerical Methods Used in Commercial Process Simulators.

A fluid is flowing in a linear (36 L ft) standard pipe Schedule 40, 1 in with a dirt factor of k = 0.0077 mm. The Reynolds number is Re = 318,000. Using the Colebrook equation calculate the friction factor using the methods listed below:

1. Solve by the Newton method to obtain the friction factor.
2. Repeat by using the Secant Method
3. Repeat by using the Direct Substitution method
4. Repeat by using the Broyden method
5. Repeat using the Wegstein acceleration method. Perform two passes before accelerating
6. Calculate the pressure drop along the linear pipe if the fluid is air that enters the pipe at {P_1=20psig, T_1=70ºF} and leaves the pipe at T_2 = 170ºF. What is the rate of heat transfer to the air?

PARAMETERS	VALUES	UNITS
Length of the pipe (L)	36	ft
Diameter (D)	1	Inch
Dirt factor (k)	0.0077	Mm
Reynolds Number	318000	

Conversions:

L = 36 ft = $36\ feet \times \frac{304.8\ mm}{1\ ft} = \ 10972.8\ mm$

D = 1 in = $1\ in \times \frac{25.4\ mm}{1in} = \ 25.4\ mm$

The friction factor can be calculated from the Colebrook equation:

$$\frac{1}{\sqrt{f}} + 1.737\ln\left\{\frac{1.256}{Re\sqrt{f}} + \frac{k}{3.707\ D}\right\} = 0 \ \bigg|\ Re = \frac{\rho v D}{\mu} > 2100$$

Let,

$$x = \frac{1}{\sqrt{f}} = \ -1.737\ln\left\{\frac{1.256}{Re\sqrt{f}} + \frac{k}{3.707\ D}\right\} = g(x)$$

$$g(x) = -1.737\ln\left\{\frac{1.256}{Re}x + \frac{k}{3.707\ D}\right\}$$

$$g'(x) = \frac{-1.737}{\left[\frac{1.256}{Re}x + \frac{k}{3.707\ D}\right]} \times \frac{1.256}{Re}$$

1. Solving by the Newton method:

The system of the Colebrook equations shows,

$$x - g(x) = 0$$

Applying the Newton method to solve the above equation we get,

$$x_{i+1} = x_i - \frac{x_i - g(x_i)}{1 - g'(x_i)}$$

Newton's Method

	x	**g(x)**	**g'(x)**	**x - g(x)**	**1 - g'(x)**
Initial Guess:	10.00	15.66	-5.66E-02	-5.66	1.06E+00
	15.36	15.38	-4.82E-02	-0.02	1.05E+00
	15.38	15.38	-4.81E-02	0.00	1.05E+00
	15.38	15.38	-4.81E-02	0.00	1.05E+00
	15.38	15.38	-4.81E-02	0.00	1.05E+00
	15.38	15.38	-4.81E-02	0.00	1.05E+00
	15.38	15.38	-4.81E-02	0.00	1.05E+00
	15.38	15.38	-4.81E-02	0.00	1.05E+00
	15.38	15.38	-4.81E-02	0.00	1.05E+00
Converging to:	15.38	15.38	-4.81E-02	0.00	1.05E+00

The newton method converges at $x = 15.38 = \frac{1}{\sqrt{f}}$ after 10 iterations,

Therefore, the friction factor is,

$$\boxed{f = 4.23 \times 10^{-3}}$$

2. Solving by the Secant Method:

To solve the equation, two initial guesses were made x_1 and x_2 for x in the Colebrook equation:

$$\frac{1}{\sqrt{x}} + 1.737 \ln\left\{\frac{1.256}{Re\sqrt{x}} + \frac{k}{3.707\ D}\right\} = g(x)$$

Using the secant step, x_3 was obtained,

$$F = x_3 = x_2 - f(x_2)\left\{\frac{(x_2 - x_1)}{(f(x_2) - f(x_1))}\right\}$$

Secant method

	x1	x2	g (1)	g (2)	F
initial guesses	0.0002	0.0003	5.694E+01	4.370E+01	6.300E-04
	0.000300	0.000630	4.370E+01	2.536E+01	0.001086
	0.000630	0.001086	2.536E+01	1.556E+01	0.001811
	0.001086	0.001811	1.556E+01	8.468E+00	0.002676
	0.001811	0.002676	8.468E+00	4.128E+00	0.003499
	0.002676	0.003499	4.128E+00	1.594E+00	0.004017
	0.003499	0.004017	1.594E+00	4.143E-01	0.004199
	0.004017	0.004199	4.143E-01	5.241E-02	0.004225
	0.004199	0.004225	5.241E-02	1.933E-03	0.004226
	0.004225	0.004226	1.933E-03	9.320E-06	0.004226

The secant method converges at $\boxed{x = 0.004225 = f}$ after ten iterations.

3. <u>Solving by the Direct Substitution method</u>

For this method, an initial guess of x was made and g(x) was calculated.
To find the second value of x,

$$x_{i+1} = g(x_i)$$

Direct Substitution Method:

	x	**g(x)**
Initial Guess:	10.00	15.66
	15.66	15.37
	15.37	15.38
	15.38	15.38
	15.38	15.38
	15.38	15.38
Converging to:	15.38	15.38

The direct substitution method converges at $x = 15.38 = \frac{1}{\sqrt{f}}$ after seven iterations,
Therefore, the friction factor is,

$$\boxed{f = 4.23 \times 10^{-3}}$$

4. Solving by the Broyden Method

For solving the Colebrook equation using Broyden method, following equation was used

$$x_{K+1} = x_K - \left(\frac{1}{1 - g'(x)}\right) * \left(x - g(x)\right)$$

Broyden Method

x	g(x)	g'(x)	x - g(x)	1 / (1 - g' (x))	dx	dG
10	1.57E+01	-0.056570884	-5.66E+00	0.946458033		
1.54E+01	1.54E+01	-0.048163127	-2.37E-02	0.95404997	5.36E+00	-2.79E-01
1.54E+01	1.54E+01	-0.048132898	-3.42E-07	0.954077485	0.0226498	-1.09E-03
1.54E+01	1.54E+01	-0.048132898	0.00E+00	0.954077486	3.267E-07	-1.57E-08
1.54E+01	1.54E+01	-0.048132898	0.00E+00	0.954077486	0	0.00E+00
1.54E+01	1.54E+01	-0.048132898	0.00E+00	0.954077486	0	0.00E+00
1.54E+01	1.54E+01	-0.048132898	0.00E+00	0.954077486	0	0.00E+00

After seven iterations the value of x converged to 15.38 and the value of $\boxed{f = 0.0042}$.

5. Solving by Wegstein acceleration method

To solve the Colebrook Equation with,

$$x_3 = x_2 - f(x_2)\left\{\frac{(x_2 - x_1)}{(f(x_2) - f(x_1))}\right\}$$

$$q = \left\{\frac{Si}{(Si - 1))}\right\}$$

Wegstein:

x1	x2	g (x1)	g (x2)	x1 - g (x1)	x2 - g (x2)	Si
10.00	20.00	1.57E+01	1.52E+01	-5.66E+00	4.83E+00	4.90E-02
20.00	15.40	1.52E+01	1.54E+01	4.83E+00	1.70E-02	4.53E-02
15.40	15.38	1.54E+01	1.54E+01	1.70E-02	-4.61E-05	4.81E-02
15.38	15.38	1.54E+01	1.54E+01	-4.61E-05	4.74E-10	4.81E-02
15.38	15.38	1.54E+01	1.54E+01	4.74E-10	0.00E+00	4.81E-02
15.38	15.38	1.54E+01	1.54E+01	0.00E+00	0.00E+00	1.00E+00
15.38	15.38	1.54E+01	1.54E+01	0.00E+00	0.00E+00	#DIV/0!

The Wegstein acceleration method converged x to 15.38 after seven iterations and the value of $\boxed{f = 0.0042.}$

6. **<u>Pressure Drop and Rate of Heat Transfer</u>**

PARAMETERS	**VALUES**	**UNITS**
Entering Pressure (P_1)	20	psig
Entering Temperature (T_1)	70	ºF
Leaving Temperature (T_1)	170	ºF

Pressure Drop is given by equation:

$$-G^2 \ln\left(x\frac{T_1}{T_2}\right) + \frac{M_W P_1^2}{2RT_a}(x^2 - 1) + G^2\ \frac{2Lf}{D} = 0$$

Where,

$$\frac{P_2}{P_1} = x$$

and G is the mass velocity (kg/ m^2 – s)

$$G = \frac{Re\ \mu}{D}$$

The viscosity (μ) found for the air is $1.963 * 10^{-5}$ kg/m-s.
Solving for G we get,

$$G = 245.7614\frac{kg}{m^2 s}$$

To solve for x, the pressure drop was goal seeked to zero by changing the value of x to get,

$$x = 0.81$$

Therefore,

$$P_2 = P_1 * x$$

$$P_2 = 20\ psig * 0.81$$

$$\boxed{P_2 = 16.2\ psig}$$

Therefore, the Pressure drop is 3.8 psig.